बैंक ऑफ़ इंडिया Bank of India BOI

भारत सरकार
इलेक्ट्रॉनिकी और सूचना प्रौद्योगिकी मंत्रालय
Government of India
Ministry of Electronics and Information Technology

सी-डैक CDAC

# Memorandum of Understanding (MoU)

For

# Quantum Financial Innovation and Emerging Technology Adoption

22nd October 2026 | 5:30 PM – 6:00 PM | MeitY, GoI, Electronics Niketan, 6, CGO Complex, Lodhi Road, New Delhi - 110003

## Overview of MoU

- Collaboration between **BOI and C-DAC** to advance **emerging technologies** for banking and financial services
- Joint **research, development, pilot implementations and technology demonstrations** in areas of mutual interest
- Exploration of **quantum technology**, **AI**, **blockchain**, and **cybersecurity** use-cases for **future-ready banking**
- Strengthening of **security, resilience, efficiency, transparency and trust** across banking operations
- Building future-ready skills through **training and capacity building** in **quantum computing, AI** and **blockchain**
- Exploration of solutions in **digital transformation**, **IoT**, **digital twin**, **cloud** and other **fintech domains**

## Key Use Cases

### Quantum Technologies for Future-Ready Banking

- Enable **quantum-safe banking** through **post-quantum cryptography** to protect against **emerging quantum threats**
- Leverage **quantum key distribution** for **secure** key exchange and **quantum-resilient communications**
- Strengthen **digital trust** through **quantum-secure digital signing** and **quantum random number generation**
- Explore **quantum computing** for **portfolio optimization, risk analytics, fraud/AML detection** and **credit assessment**
- Conduct **quantum readiness assessments** to identify **vulnerabilities** and prepare the bank for the **quantum era**

### AI for Intelligent Banking

- Leverage **generative AI** and **agentic AI** for **customer service**, **document processing** and **compliance automation**
- Run pilots across **fraud detection, risk assessment, financial analytics, personalized banking** and **decision support**
- Implement AI in a **secure**, **privacy-aware** and **responsible** manner, with built-in **explainability** and **transparency**

### Blockchain for Trusted Banking

- Evaluate blockchain applications for **digital transactions, smart contracts, trade finance, KYC and digital identity**
- Facilitate secure **inter-bank reconciliation and settlement, asset tokenization** and **data sharing**
- Enhance **fraud prevention** and enable **tamper-evident audit trails** for greater **transparency** and **trust**